

ISYS2101 – Software Engineering Project Management

GreenRoute HCMC Final Report

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I declare that in submitting all work for this assessment I have read, understood and agreed to the content and expectations of the Assessment declaration.

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Executive Summary

This report presents the completed GreenRoute HCMC project, a cross-platform EV-focused route guidance system delivered through web browsers and Android WebView. The project addresses a practical limitation of conventional navigation tools: EV users do not only need the shortest or fastest route, but also need to know whether a trip is feasible with their remaining battery, whether a charging or battery-swap stop is required, and how local air-quality conditions may affect route choice. The final system therefore combines route planning, battery-feasibility checking, charging guidance, AQI visualisation, vehicle profile management, and local trip history into a single route-planning workflow.

The project is locally relevant because Vietnam has committed to achieving net-zero emissions by 2050 and Ho Chi Minh City is pursuing greener transport initiatives, including electrification plans for buses and motorbike-based services [1] [2]. This context makes EV-oriented navigation more than a theoretical concept, especially for urban electric motorbike and scooter users who travel in dense, stop-and-go conditions. GreenRoute HCMC responds to this need by focusing on a pilot area in Ho Chi Minh City while keeping the system architecture extensible for future expansion.

The completed MVP achieved the main objectives set in the original proposal. Users can select or create an EV profile, enter a battery percentage, search for a route, compare Optimal, Alternative, and Lower-Pollution route options, view AQI information, and receive charging or battery-swap recommendations when the destination is not safely reachable. The system also follows a privacy-conscious design by storing vehicle profiles, route history, and cached datasets locally on the user's device or browser rather than relying on central storage of personal trip data.

During development, the team made several practical adjustments to protect the MVP scope and delivery schedule. The Android implementation was completed through WebView rather than a separate native Android feature layer, allowing the same frontend and JavaScript route-planning logic to be reused across platforms. Weather integration was reduced from a dedicated route-calculation feature to contextual information in AQI/location popups, and route differentiation was adjusted because many Ho Chi Minh City route alternatives were highly similar in practice. The team also responded to backend schedule delays by reallocating work, using prepared mock data, and accelerating other integrations so frontend and backend progress could continue in parallel.

Overall, GreenRoute HCMC was successfully delivered as a working MVP that demonstrates EV-aware navigation, battery-based trip feasibility, forward-progress charging guidance, AQI-aware route scoring, cross-platform access, and local privacy-conscious storage. Testing confirmed that the main user flows operated across desktop browsers, mobile browsers, and Android WebView, while minor limitations such as keyboard navigation gaps, WebView text-selection behaviour, and network-recovery handling were documented for future improvement.

1. Project Description

Problem Statement

Traditional navigation applications are designed to optimise travel time or distance, which works well for most drivers but falls short for electric vehicle (EV) users. EV riders must also account for remaining battery

capacity, estimated driving range, charging or battery-swap availability, and whether their destination is realistically reachable without running out of power.

This is particularly relevant in Ho Chi Minh City, where electric motorbikes and scooters are increasingly common for commuting and delivery. Congested, stop-and-go urban conditions make battery management more critical, and in practice, a route is only useful to an EV user if it is achievable with the available charge and supported by accessible charging infrastructure.

Ho Chi Minh City is also actively promoting greener transport and reduced pollution through its electrification initiatives. While separate services already exist for navigation, charging locations, air-quality data, and battery monitoring, these are spread across different platforms, forcing users to switch between multiple apps before making travel decisions.

This project addresses that gap by combining EV-aware route planning, charging guidance, and pollution-awareness into a single cross-platform system, supporting more practical travel decisions for EV users while offering environmentally conscious routing for general users as well.

Product Outcome

The final product is GreenRoute HCMC, a cross-platform EV route-guidance system accessible via web browsers and Android devices. The Android version uses WebView integration, allowing the same frontend and route-planning logic to be shared across both platforms.

The system allows users to search for routes, configure EV vehicle profiles, estimate remaining travel range from current battery percentage, and determine whether a destination is directly reachable. Users can also compare multiple route options, view AQI overlays with weather information and lower-pollution alternatives, receive charging or battery-swap recommendations when battery is insufficient, and store route history and vehicle data locally on their device.

The system integrates the following external services: OSRM for routing, Nominatim for geocoding, IQAir for AQI data, Open-Meteo for weather context, and Leaflet with OpenStreetMap/CARTO for map visualisation.

The completed MVP focuses on practical EV travel support within Ho Chi Minh City, with an extensible architecture designed for future expansion.

Deviations from Original Proposal

Although the final implementation achieved the main project objectives, some changes were made during development due to technical practicality, and implementation discoveries.

Routing implementation scope

The initial concept explored more advanced route differentiation and environmental scoring. In practice, many routes within Ho Chi Minh City were highly similar due to dense urban road layouts and evenly distributed AQI conditions.

Weather integration scope reduction

While the original proposal included a dedicated weather overlay with temperature-based range adjustments, weather information is currently surfaced as contextual data within AQI and location popups. This scoped implementation keeps the feature present and accessible without introducing the complexity of real-time calibration into route and battery calculations, leaving it as a natural candidate for future enhancement.

2. Product Design

GreenRoute HCMC is built on Node.js/Express and supports desktop browsers, mobile browsers, and Android WebView. Rather than developing a separate native Android feature layer, the mobile version reuses the same interface and JavaScript route-planning logic served through WebView. This kept the design and functionality consistent across platforms while avoiding the overhead of duplicating features as native Android components.

2.1. Product's architecture

GreenRoute follows a hybrid client-server architecture, with a Node.js/Express backend and most of the interactive route-planning experience running in the browser. This was a practical choice given that the map interface, vehicle settings, AQI display, battery input, route cards, and charging-stop markers all require fast responses to user actions. The Android app reuses the same frontend and backend logic via WebView rather than maintaining separate implementations.

Local storage is central to the final design. Vehicle profiles and route history are stored on the user's device, while AQI and charging datasets are cached in the browser's IndexedDB. This supports the privacy-conscious approach from the proposal, users retain full control over their data, and nothing sensitive such as search history, past trips, or vehicle profiles is collected externally.

The architecture is therefore divided into four major layers: user access through desktop browser, mobile browser, or Android WebView; a shared frontend layer for map interaction and route-result display; a browser-side planning and cache layer for vehicle profiles, route history, AQI/charging datasets, and battery-feasibility logic; and a Node.js/Express backend layer for routing/data endpoints and static asset delivery. Appendix B maps these architectural components to their implementation files/modules.

2.2. Key features

The final product focuses on the features that are most important for demonstrating EV-aware routing: route search, battery feasibility, charging or battery-swap guidance, AQI awareness, and cross-platform access.

2.2.1 Vehicle profile and trip setup

GreenRoute allows users to set up their trip based on the EV they are using. Users can add or select a vehicle profile with details such as brand, vehicle type, battery percentage, driving range, and supported charging option. The selected vehicle information is then used together with the battery slider in the direction panel to estimate whether a planned trip is realistic.

This feature was important because the project is not only about showing a route from one place to another. Since the application is designed for EV users, the route result needs to consider whether the vehicle has enough battery to complete the trip safely.

User interface reference: Vehicle menu/modal, selected vehicle label, and battery slider in the Direction panel.

Contribution to the problem: This gives the route planner the basic vehicle and battery information needed to provide EV-aware guidance instead of only showing a normal shortest or fastest route.

2.2.2 EV-aware route planning and map visualisation

This is the main feature of the web application. Users can search for a starting point and destination, view the selected locations on the Leaflet map, and generate route options shown through coloured route lines and route information cards. The map interface supports current-location input, start and destination markers, and route selection directly from the displayed results.

During implementation, the route planner was developed to do more than only display directions. After a route is generated, the system estimates whether the trip can be completed using the selected vehicle's current battery level and driving range. The route card shows practical feasibility information, such as whether the destination is directly reachable, whether the arrival battery may be dangerously low (less than 15%), whether a charging or battery-swap stop is needed, or whether no safe energy-stop chain is available.

The product also supports route comparison through options such as Optimal Route, Alternative Route, and Lower-Pollution Route. These options are only useful when they are meaningfully different, so the current implementation focuses on showing practical route candidates rather than forcing duplicate or unrealistic choices. For example, Lower-Pollution Route often overlaps with Optimal Route or Alternative Route, even with the modifier that treats more polluted paths as longer (the weight can theoretically get up to 40%, but as pollution tends to be evenly dispersed, routes commonly differ by only 5%.) This keeps the route comparison useful while still prioritising battery feasibility and travel practicality.

When the destination is not safely reachable, GreenRoute recommends a battery charging/swapping stop as part of the route-planning process. The recommendation is based on factors such as whether the station is within the remaining battery range, whether it helps the user continue moving toward the destination, the station type, and vehicle compatibility. This makes the stop recommendation more journey-aware instead of simply listing the nearest charger.

User interface reference: Vehicle menu/modal, selected vehicle label, and battery slider in the Direction panel.

Contribution to the problem: This gives the planner the battery and range assumptions needed to judge whether a route is actually usable for the rider.

2.2.3 AQI overlay and lower-pollution scoring

The Lower-Pollution Route option above utilises AQI data, which is provided by IQAir. This data is scraped from their publicly available resources on their domain instead of available API as due to its limited feature (can only provide AQI of user's current location.) Users can view AQI information on the map using "Air Quality" overlay, this makes air-quality conditions visible inside the same interface used for route planning. Because the AQI data available on IQAir includes third-party and non-government contributors, the lower-pollution route should be understood as an estimated lower-exposure option rather than a guaranteed safe-air route.

User interface reference: Air Quality map toggle, AQI legend, AQI markers/heatmap, Avg AQI, and Pollution Factor in route cards.

Contribution to the problem: This connects the route-planning feature with HCMC's air-quality issue, helping users consider pollution exposure while still keeping the route practical for EV travel.

2.2.4 Weather context in AQI popups

Open-Meteo weather data is shown as contextual information in AQI/location popups, including temperature, humidity, wind, pressure, and weather description. This is not used to recalculate battery range or route scores.

User interface reference: AQI station popup.

Contribution to the problem: This was mentioned as an optional feature in the initial report. It provides users with extra information to be aware of the weather conditions along their trip.

2.2.5 Local history, caching, and data freshness

The application stores recent route history locally and caches AQI/charging datasets in the browser. A data-freshness variable helps refresh missing or stale datasets before route planning when needed.

User interface reference: History panel and data-refresh prompt/gate before planning.

Contribution to the problem: This supports repeated use of the app and keeps personal route/vehicle data local while still aligning with Privacy-conscious direction.

2.2.6 Cross-platform access through shared web implementation

Compared with the initial proposal, the final implementation still follows the cross-platform direction, but the Android version is handled through WebView rather than a separate native Android feature layer. This became a more practical approach during development because the same mobile interface and JavaScript route-planning logic could be reused across platforms, keeping the system consistent while reducing duplicated work.

User interface reference: Desktop page, mobile page, and Android WebView loading the same hosted GreenRoute app.

Contribution to the problem: Users can conveniently use this app on browser or as a downloaded application to assist them in everyday travelling with their EV.

3. Project Management

Schedule Analysis

This section will showcase the difference between the original proposed schedule during the planning phase, compared to the final schedule, which was adapted to better suit the project as it evolves over time. Below are the major differences between the two schedules, for a fully detailed look into both variations, refer to the following [link](#).

Alteration 1: The development and integration of the battery awareness feature for routing were rescheduled from Week 3 to Week 5 due to the limited availability of the project's main backend developer, who was responsible for implementing the component. In addition, several technical issues affecting other critical parts of the system required additional time for debugging and adjustments, resulting in development priorities being temporarily shifted. To minimise the impact of the delay, the team accelerated the implementation of the weather and local storage integrations. Furthermore, mock data prepared during Weeks 1 and 2, as originally planned, enabled frontend developers to continue developing and testing their components independently without being blocked by backend progress.

Alteration 2: As mentioned in Alteration 1, the development of the local storage and weather integrations was accelerated to compensate for the delayed progress of one backend component. By implementing this shift, the other backend developer was able to continue working with minimal downtime, without needing to wait for the progress of the first backend developer.

Alteration 3: Additional technical improvements were proposed during Week 6, after the team had stabilized following the schedule adjustments in the previous weeks. As the team had progressed ahead of the planned timeline, there was additional capacity to introduce new features aimed at further refining and enhancing the final product. These improvements primarily focused on optimizing routing efficiency, thereby improving the overall user experience.

Alteration 4: Preparation for the mid-term project presentation was moved from Week 4 to Week 5. After considering the number of critical tasks scheduled for Week 4, we decided to prioritize completing those tasks before beginning any presentation preparation. Additionally, after re-evaluating the requirements and purpose of the presentation, we determined that it was not necessary to begin preparation earlier.

Performance Evaluation

After accounting for the schedule adjustments and project changes throughout development, the team successfully achieved a completion rate of 100%, with all predefined tasks completed within the project scope. Although the project experienced its greatest deviation from the original proposed schedule between Weeks 3 and 5, the revised scheduling strategy allowed the team to maintain overall progress in line with the originally planned timeline.

The schedule modifications implemented during this period helped minimise development downtime and ensured that team members could continue working productively despite delays affecting several backend components. By redistributing development priorities and accelerating other planned integrations, the team was able to maintain consistent progress across both frontend and backend development.

From Week 5 onwards, after the revised schedule had stabilised, development progress accelerated significantly. This allowed the team not only to recover from earlier delays, but also to progress ahead of the original schedule in later stages of the project. The improved pace of development also created additional opportunities to introduce technical refinements and optimisations aimed at enhancing the final product.

Throughout the project, regular team meetings were conducted to monitor development progress, discuss technical challenges, review completed tasks, and coordinate upcoming deliverables. Meetings were held both online and in person depending on team availability.

The meeting records included discussion summaries, task allocations, progress updates, identified issues, and planned actions for subsequent development periods. These meetings played an important role in supporting schedule adjustments, redistributing workload when necessary, and maintaining communication between frontend and backend development teams.

For a more detailed breakdown of the completion rates across the various stages of the project, and a look into the details of meetings, please refer to the following [Google Sheet](#).

Success Criteria Review

This section presents an evaluation of the final project outcome against the original success metrics established during the planning phase. Overall, the team was able to meet all the success criterias originally laid out. For full details of all the success metrics, please refer to Appendix C.

4. Expected Outcome

The final outcome of this project is GreenRoute HCMC, a working cross-platform EV-aware route guidance system accessible through both desktop/mobile web browsers and an Android application via WebView integration. The product successfully demonstrates how EV-specific navigation support can be integrated into a practical route-planning interface tailored to urban travel conditions in Ho Chi Minh City.

Description of all required functions

EV-aware route planning

Users can enter an origin and destination, select their EV model, and specify the current battery percentage before planning a route. The system estimates whether the trip is realistically reachable using the vehicle's configured driving range and current battery state.

The planner classifies trip feasibility into practical categories such as:

- ❖ Directly reachable
- ❖ Near battery limit
- ❖ Charging stop required
- ❖ No feasible energy-stop chain available

Interactive map visualisation

The application provides an interactive map interface implemented using Leaflet and OpenStreetMap featuring:

- ❖ Locations searching
- ❖ Current-location detection
- ❖ Route geometry view
- ❖ Multiple route candidates comparison
- ❖ Charging stations and AQI overlays displayed

Multi-route comparison

GreenRoute provides three selectable route types:

- ❖ Optimal Route
 - Main routing logic based on OSRM (Open Source Routing Machine) for OpenStreetMap data.
- ❖ Alternative Route
 - Main routing logic based on OSRM (Open Source Routing Machine) for OpenStreetMap data.
- ❖ Lower-Pollution Route
 - Applies AQI-based weighting to polluted segments balancing environmental awareness with realistic EV travel constraints.

Charging and battery-swap recommendation

When a destination is not safely reachable, the system searches for charging or battery-swap stations that:

- ❖ Are reachable within remaining battery range
- ❖ Continue progress toward the destination
- ❖ Match vehicle charging compatibility

AQI overlay and pollution-aware guidance

GreenRoute integrates AQI information scraped from IQAir datasets and displays it directly on the map interface through overlays and popup information. AQI data contributes to the Lower-Pollution Route scoring system, allowing users to select routes with lower estimated pollution exposure where practical.

Vehicle profile management

Users can create and manage EV profiles containing:

- ❖ Vehicle brand/model
- ❖ Vehicle type
- ❖ Battery percentage
- ❖ Estimated maximum range
- ❖ Charging compatibility
- ➔ This information is reused during route planning and feasibility estimation.

Local trip history and caching

Recent trips, Vehicle information, Cached AQI data, Cached charging datasets are stored inside browser localStorage rather than on a centralised server. This supports the project's privacy-conscious design direction while also reducing repeated API/data requests.

Cross-platform support

The final implementation supports:

- ❖ Desktop browsers
- ❖ Mobile browsers
- ❖ Android WebView deployment
- The Android version reuses the same frontend interface and JavaScript logic rather than maintaining a separate native implementation, reducing duplicated development effort while keeping the experience consistent across devices.

Evidence of creativity/innovation

Battery-feasibility focused navigation

This product demonstrates creativity primarily through its focus on practical EV travel rather than conventional shortest-path navigation. Unlike most traditional navigation systems, GreenRoute evaluates whether a route is realistically completable with the user's current battery state before presenting guidance. If the remaining battery is lacking, the system automatically suggests a route with charging points along the way toward the destination.

AQI-integrated route scoring

The project combines route planning with environmental-awareness by incorporating AQI weighting into route selection. While pollution-aware routing exists in some research systems, integrating it directly into an EV-focused navigation workflow for HCMC provides a more locally relevant implementation.

Privacy-conscious local storage design

Unlike many modern navigation systems that centrally collect trip history and user data, GreenRoute stores route history and vehicle information locally on the user's device/browser. This reduces privacy concerns while excluding the need to build a database.

Shared cross-platform implementation

Rather than developing completely separate Android and web logic, the project reused frontend interfaces, JavaScript planning logic from web through Android WebView integration and shared backend APIs. This

significantly reduced duplicated development complexity while maintaining feature consistency across platforms.

Testing and deployment status

Testing for GreenRoute HCMC focused on verifying the functionality, and cross-platform behaviour of the application across desktop browsers, mobile browsers, and Android WebView environments. A structured testing spreadsheet was used to record test cases, expected behaviour, actual outcomes, and identified issues throughout development.

[Testing Spreadsheet](#)

The testing process included:

- ❖ Route planning and map rendering
- ❖ Vehicle profile creation and battery-feasibility estimation
- ❖ Charging and battery-swap recommendation
- ❖ AQI overlay display and lower-pollution routing
- ❖ LocalStorage and IndexedDB persistence
- ❖ Mobile responsiveness and Android WebView compatibility
- ❖ End-to-end user flows and error handling

Testing was primarily conducted using Google Chrome, Mozilla Firefox, and Android mobile devices. Most major functions operated successfully across supported environments, including route generation, AQI overlays, vehicle management, and local trip-history storage. The application also maintained consistent behaviour between desktop and mobile layouts through the shared frontend implementation.

Several minor issues were identified during testing. These included incomplete keyboard tab navigation support, temporary text-selection behaviour during double-tapping in Android WebView, and a network-recovery issue where route planning could become stuck if connection loss occurred mid-request. These limitations were documented for future improvement but did not prevent the core MVP functionality from operating correctly.

Overall, testing confirmed that the system successfully met the primary project objectives for EV-aware route guidance, charging-stop recommendation, pollution-aware routing, and cross-platform accessibility.

Hosting

GreenRoute HCMC is deployed as a live web application and can be accessed through standard desktop and mobile web browsers. The same hosted application is also loaded inside the Android version through WebView integration, allowing both platforms to use the same backend services and route-planning logic.

Deployment location

For links to the GreenRoute website, GreenRoute APK file, and GreenRoute Project Repository, please refer to the table Appendix A.

Hosting architecture

The application is hosted using a Node.js and Express server environment. The deployment includes:

- ❖ Frontend web pages for desktop and mobile layouts
- ❖ Backend API endpoints for routing and data access
- ❖ AQI and charging-data refresh services
- ❖ Static asset hosting for JavaScript, CSS, and map components

The Android version accesses the same hosted application through Android WebView, allowing consistent functionality between platforms without maintaining separate backend infrastructure.

Environment requirements

To access and evaluate the product, the following environment is recommended:

- ❖ Web version
- ❖ Modern browser such as:
 - ❖ Google Chrome
 - ❖ Microsoft Edge
 - ❖ Firefox

Android version

- ❖ Android device with WebView support enabled
- ❖ Internet connection required for API, data update and retrieval

Login and access instructions

The current MVP version does not require mandatory user authentication. Users can directly access the route-planning interface after opening the application.

To evaluate the main functionality:

- ❖ Open the web application or Android version
- ❖ Add or select a vehicle profile

- ❖ Enter current battery percentage
- ❖ Choose a start location and destination
- ❖ Generate route options
- ❖ Review:
 - Battery-feasibility results
 - Charging-stop recommendations
 - AQI overlays
 - Route comparison cards

Trip history and vehicle information are stored locally on the user's device/browser rather than on a central server.

Known deployment limitations

Because the project is implemented as an MVP prototype:

- ❖ Some external data services may occasionally experience rate limits or temporary unavailability
- ❖ AQI and charging datasets depend on third-party sources
- ❖ Performance may vary depending on network quality and device capability
- ❖ The Android version currently uses WebView rather than a fully native Android implementation

Product demonstration

Link to the demonstration video below: https://www.youtube.com/watch?v=aa_4ggNABEM

5. Conclusion

Summary of Key Findings and Achievements

The GreenRoute HCMC project delivered a cross-platform electric vehicle (EV) route-guidance system designed to bridge the gap between conventional shortest-path navigation and the real-world requirements of EV travel in urban Vietnam. The final solution runs in both desktop and mobile web browsers and is also packaged as an Android application through WebView integration, enabling the web and Android versions to reuse the same backend services and route-planning logic.

Over the development period, the team completed 100% of the predefined tasks and met all thirteen success metrics established during the planning phase. Key functional outcomes include battery-aware route planning that categorises trip feasibility as directly reachable, near limit, charging-stop required, or infeasible; a three-route comparison featuring Optimal Route, Alternative Route, and Lower-Pollution Route options; forward-direction charging or battery-swap recommendations that prioritise stations supporting continued travel toward the destination rather than simply the closest by distance; an AQI overlay that incorporates IQAir data into both the map interface and the Lower-Pollution Route scoring model; and a vehicle-profile

module that lets users set brand, model, driving range, battery state, and charging compatibility for the planned trip.

The system also met its non-functional commitments. A privacy-oriented local-storage approach retains trip history and vehicle data on the user's device via `localStorage` and `IndexedDB`, avoiding centralised collection of personal data while still supporting repeated use. Cross-platform consistency was maintained by reusing a single frontend and JavaScript planning layer across web and Android, which kept the experience consistent and reduced duplicated implementation effort. Overall, these outcomes position GreenRoute as a locally relevant contribution to Ho Chi Minh City's green-transport agenda and as a practical demonstration that EV-aware navigation can be built using openly available routing, geocoding, and air-quality services without requiring a fully bespoke ecosystem.

Reflections on Project Management

Project delivery underscored the importance of adaptive planning rather than strict compliance with the original timeline. The most significant disruption occurred in Week 3, when the primary backend developer's limited availability delayed the battery-awareness feature until Week 5. Instead of allowing progress to stall, the project manager and technical leads reallocated tasks and brought forward the weather and local-storage integrations that had initially been scheduled for later. This adjustment was enabled by the mock data produced in Weeks 1 and 2, which reduced frontend reliance on backend readiness and illustrated how early investment in interface definitions and test fixtures can produce outsized benefits later in the delivery cycle.

Maintaining scope discipline further supported recovery. Postponing preparation for the mid-term presentation from Week 4 to Week 5 reflected a practical assessment that implementation work should not be displaced by presentation preparation, which could be completed nearer to the presentation deadline. From Week 5 onward, the team progressed ahead of the revised schedule, creating additional capacity for routing-efficiency optimisation and other refinement tasks that were not explicitly included in the original proposal. In addition, the cross-platform choice to reuse the web frontend via Android `WebView`, rather than developing a separate native feature layer, freed substantial engineering effort at a critical stage of the project.

Communication practices reinforced these outcomes. Hybrid meetings held online or in person depending on availability facilitated task allocation, surfaced integration challenges early, and maintained alignment between frontend and backend development during the most unstable mid-project period. The proposal's risk register also operated as an active management tool rather than a static planning artefact: the human, estimation, and integration risks identified during planning closely matched the issues encountered during execution. Overall, the team's readiness to respond quickly to emerging signals, rather than treating the initial plan as fixed, was a key factor in achieving the final completion outcome.

Recommendations for Future Projects

A number of actionable recommendations can be drawn from the delivery experience of GreenRoute HCMC. First, knowledge redundancy should be established at project commencement by ensuring that at least two team members maintain practical familiarity with each critical module. Although the Week 3 disruption was mitigated, this outcome was largely attributable to an architecture that enabled parallel development across relatively independent components; a more tightly coupled subsystem would likely have resulted in a longer schedule impact. Accordingly, cross-training and shared technical documentation should be planned as recurring activities rather than relying on ad hoc handovers.

Second, integration testing should be initiated earlier in the timeline instead of being concentrated near deployment. Establishing explicit application programming interface (API) contracts prior to parallel implementation would reduce the likelihood of late-stage integration issues, particularly across the frontend route renderer, backend dataset endpoints, and external services such as OSRM and Nominatim. Third, automated testing should be incorporated earlier in the development lifecycle. The existing verification approach depended heavily on manual user-interface walkthroughs, which do not scale effectively and offer limited regression assurance; therefore, introducing automated end-to-end testing (e.g., via Playwright), complemented by unit tests for AQI scoring and battery-feasibility logic, would help prevent undetected regressions in future iterations.

Fourth, several ancillary practices observed during the project would benefit from formal standardisation. Centralising scraper scripts under clearly documented npm commands, replacing the IQAir scraper with a maintained data feed when available, and providing a sample environment-configuration file would improve maintainability without changing externally visible system behaviour. In addition, maintaining a structured decision log to capture the rationale behind key trade-offs (for example, the choice of a WebView-based approach over a native implementation) would allow future teams to reassess decisions with complete context rather than inferring intent from commit history.

Finally, subsequent iterations should evaluate a native Android implementation to enable advanced functionality such as offline routing and on-device sensor integration, while continuing to apply the staged scope discipline (minimum viable product first, followed by enhancements only once the core is stable) that proved effective during this project. Collectively, these recommendations seek to retain GreenRoute HCMC's demonstrated strengths: adaptive planning, parallel work-stream design, and locally relevant feature scoping while directly addressing the specific challenges encountered during delivery.

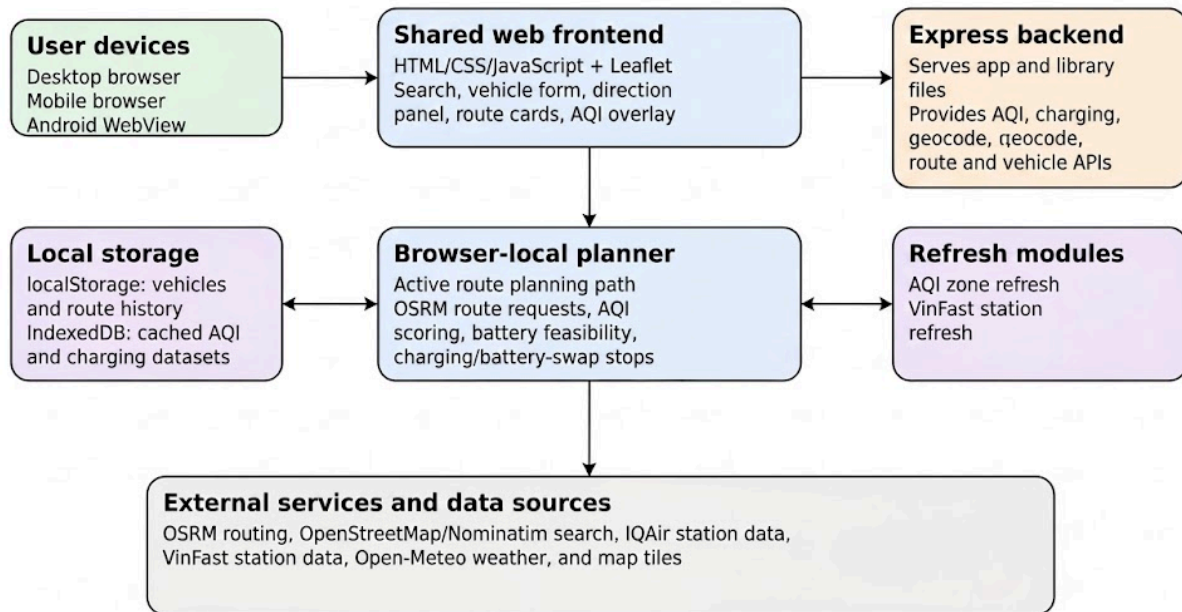
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APPENDIX A: Hyperlinks

Link	Description
GreenRoute Demonstration Video	Demonstration video showcasing the GreenRoute Website, and the GreenRoute Android Application
GreenRoute Website	GreenRoute website, hosted on Render
GreenRoute APK	GreenRoute APK file of the Android application
Github Repository	GreenRoute project repository
Project Schedule Analysis	Analysis of GreenRoute project schedule
 GreenRoute_UI_Test_Cases	GreenRoute testing documentation
 SEPM - GreenRoute - Meetings & ...	Meetings and progress tracking

APPENDIX B: Components Diagram



Component	Role(s)	Corresponding files/modules
Shared web/mobile frontend	Main user interface: Leaflet map, search bar, direction panel, route cards, vehicle modal, AQI overlay, and history panel.	public/pages/index-desktop.html, public/pages/index-mobile.html, public/js/map.js, menu.js, routes.js, pollution.js
Browser-local route planner	Active planning path for the current desktop/mobile pages. It requests route data, compares route options, scores AQI exposure, estimates battery arrival, and adds charging or battery-swap detours where needed.	lib/logic/client_route_planner.js (specifically window.greenRouteClientPlanner.planRouteWithLocalData which is called by routes.js)
Local storage and cache layer	Stores vehicle profiles and route history locally, and caches AQI/charging datasets to reduce repeated downloads and support freshness checks.	lib/logic/browser_data_store.js and lib/logic/local_db_control.js controls localStorage and IndexedDB

Node.js/Express backend	Serves the application and library files, chooses desktop/mobile pages, exposes data APIs, geocoding support, vehicle CRUD endpoints, and a server-side route-planning endpoint.	index.js
Data refresh and normalisation modules	Refresh and normalise AQI zone data and VinFast charging/battery-swap station datasets before they are returned to the app.	lib/scripts/aqi_live_refresh.js, scraper.js, vinfast_station_refresh.js, charging_live_refresh.js; lib/logic/aqi_data_logic.js and charging_data_logic.js
External services and data sources	Provide map tiles, geocoding, route geometry, AQI readings, charging/battery-swap station data, and contextual weather information.	OSRM, Nominatim, Open-Meteo, Leaflet/CARTO tiles. Vinfast stations and IQAir are scraped via Playwright.

A typical user flow begins with the user opening the route interface, selecting or adding a vehicle, entering the current battery level, and choosing a start and destination. Location search is handled via the Nominatim API. The frontend then passes the start and end coordinates, battery percentage, vehicle range, and vehicle type into the route planner. The planner loads fresh or cached AQI and charging datasets, requests route geometry from OSRM, estimates battery feasibility, and where a direct trip is not feasible, ranks suitable forward charging or battery-swap stations. AQI data drives both the Lower-Pollution Route scoring and overlay display, while weather data from Open-Meteo is currently surfaced only within AQI and location popups. Final route options are rendered as coloured map lines with accompanying route cards, and the completed trip is saved to localStorage.

APPENDIX C: Success Metric Table

SUCCESS METRIC COMPARISON TABLE				
Success Metric		Fully Met	Partially Met	Not Met
Schedule	The project is completed within the approved capstone timeline.	X		
	Major milestones for analysis, design, implementation, testing, and demonstration are completed on schedule.	X		
Scope	The final system includes all defined MVP features.	X		
Quality and reliability	All core user flows pass planned functional tests.	X		
	No unresolved critical defects remain at final testing.	X		
	Battery feasibility classification works correctly in benchmark scenarios covering directly reachable trips, near-limit trips, and trips that require a charging stop.	X		
	In benchmark test scenarios where the destination is not directly reachable, the system recommends a charging or battery-swap point that is both reachable and located within the defined forward travel corridor toward the destination.	X		
Performance	The system returns route guidance results within an acceptable response time of 5 seconds under normal demo conditions in the pilot zone.	X		
	Core map interactions and trip planning screens operate smoothly on a standard Android device.	X		

User effectiveness and satisfaction	Users can successfully complete key tasks such as planning a route, checking battery feasibility, and following a recommended charging stop without assistance.	X		
	User feedback indicates that the system is understandable, practical, and useful for EV travel planning.	X		
	Supervisors and testers agree that the final system clearly demonstrates a realistic solution to the stated problem.	X		
	Users are able to understand the trip-feasibility result and distinguish between directly reachable trips, near-limit trips, and trips requiring a charging stop.	X		